BIO60I - Proteomics MCQS

Proteomics (Virtual University of Pakistan)



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BIO601

Chapter 1: Proteomics and the New Biology

1. **Who coined the terms "proteomics" and "proteome"?**

2. a) John Yates
3. **b) Marc Wilkins**
4. c) Daniel Liebler
5. d) Patrick Brown
6. **Explanation:** Marc Wilkins introduced these terms in the early 1990s to parallel “genomics” and “genome”.

7. **Which of the following is NOT a key development leading to the new biology?**

a) Genome-sequencing projects

b) Bioinformatics tools

c) PCR and hybridization

d) RNA interference

Explanation: RNA interference is important but not one of the three foundational tools mentioned in this chapter.

8. **What advantage do microarrays provide over traditional Northern blots?**

a) Measure protein levels

b) Assess thousands of gene expressions simultaneously

c) Detect posttranslational modifications

d) Analyze protein-protein interactions

Explanation: Microarrays allow parallel gene expression analysis.

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9. Why are protein levels not always predictable from mRNA levels?

a) mRNA is hard to isolate

b) Differences in translation efficiency and protein stability

c) DNA contamination

d) Post-splicing errors

Explanation: mRNA stability, translation, and protein turnover vary independently.

10. What is the main analytical challenge of proteomics compared to genomics?

a) Data storage

b) Lack of amplification tools like PCR

c) Absence of databases

d) Complexity of DNA sequences

Explanation: Proteins cannot be amplified like nucleic acids.

11. Which is NOT one of the four essential tools of proteomics?

a) Mass spectrometry

b) Protein-separation technologies

c) CRISPR gene editing

d) Bioinformatics databases

Explanation: CRISPR is unrelated to proteomics analysis.

12. Which technique offers peptide mass and sequence analysis in proteomics?

a) PCR

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b) Mass spectrometry

c) Electrophoresis

d) Western blot

Explanation: MS is the backbone of protein ID via peptide data.

13. Which application is NOT part of proteomics?

a) Mining proteomes

b) Protein expression profiling

c) DNA sequencing

d) Mapping protein modifications

Explanation: DNA sequencing is a genomics task.

Chapter 2: The Proteome

9. **Why is the proteome considered more complex than the genome?**
   a) It contains RNA too
   b) It mutates rapidly
   c) Proteins exist in multiple posttranslationally modified forms
   d) It's less stable
   Explanation: Each gene can produce several modified protein forms.
10. **What process adds phosphate groups to proteins for regulation?**
    a) Glycosylation
    b) Phosphorylation
    c) Acetylation
    d) Methylation
    Explanation: Phosphorylation is a common reversible modification.
11. **Which of the following is a modular unit in proteins responsible for specific functions?**
    a) Promoter
    b) Domain

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c) Exon
d) Intron
Explanation: Domains and motifs give functional properties to proteins.

12. **What percentage of the human genome encodes proteins of unknown function?**
a) ~10%
b) ~20%
c) ~40%
d) ~80%
Explanation: Roughly 40% remains uncharacterized functionally.

13. **What are paralogs?**
a) Genes from different species
b) Duplicated genes with similar sequences within the same organism
c) Noncoding RNAs
d) Pseudogenes
Explanation: Paralogs are similar gene copies within a genome.

14. **Codon bias refers to:**
a) Mutation rate in genes
b) GC content in coding regions
c) Preference for certain codons over others encoding the same amino acid
d) Protein folding efficiency
Explanation: Codon bias influences translation efficiency.

15. **Which of the following factors affects the protein level in a cell?**
a) Chromatin structure only
b) Epigenetics only
c) Transcription rate, translation efficiency, and protein degradation
d) DNA replication rate
Explanation: Protein levels are controlled at multiple expression levels.

Would you like **more MCQs chapter-wise** or in **bulk (e.g., 50 or 100)**?

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